

Term Projects – Data Analytics Course

General Instructions:

- You may select one of the suggested project titles below or propose your own.
- Teams: Up to **3 students per team**.
- Projects should involve **data collection, preprocessing, analysis, visualization, and insights**.
- Creativity, relevance to your background discipline, and entrepreneurial potential are encouraged.

Suggested Project Titles

1. Startup Success Prediction Using Investment and Market Data
2. Customer Churn Analytics for Subscription-Based Businesses
3. Market Trend Forecasting with Social Media and News Data
4. Competitive Intelligence Analytics Using Web & Public Data
5. Social Media Sentiment Analysis for Brand Monitoring
6. Cybersecurity Threat Detection Using Network Traffic Logs
7. Predicting Construction Delays Using Project Management Data
8. Smart City Traffic Flow Analysis with Sensor and GPS Data
9. Energy Consumption Analytics in Green Buildings
10. Structural Health Monitoring with IoT Sensor Data
11. Predictive Maintenance for Industrial Machines Using Sensor Data
12. Optimization of Supply Chain Logistics Using Data Analytics
13. Wearable Devices Data for Stress and Sleep Analysis
14. Predicting Hospital Readmission Rates Using Patient Records
15. Analytics for Healthy Lifestyle Recommendation from Fitness Data
16. Text Mining of Health Forums for Public Health Insights
17. Fraud Detection in Online Transactions Using Machine Learning
18. Customer Behavior Analysis Using Retail Transaction Data
19. Real-Time Event Detection from Twitter Streaming Data
20. Spatial Data Analytics for Predicting Landslide Risk Zones
21. Geostatistical Modeling of Groundwater Quality Using Sensor Data
22. Drone-based Photogrammetry Data Analytics for Construction Site Monitoring
23. Seismic Data Analytics for Earthquake Pattern Detection
24. Power Consumption Forecasting Using Smart Meter Data Analytics
25. Fault Detection in Electrical Grids Using Machine Learning

(Students are free to propose their own projects if they have a strong idea relevant to their field.)

Project Development Workflow – 5 Phases (Biweekly Deliverables)

The project follows a **Data Science Workflow**. Each phase must be submitted/demonstrated **every 2 weeks**.

Phase 1 – Problem Definition & Data Source Identification

- Define the research problem clearly.
- Identify project goals (prediction, classification, clustering, visualization, etc.).
- Select appropriate data sources (internal, external, APIs, sensors, etc.).
- Deliverable: 2–3 page project proposal with problem statement, objectives, and data sources.

Phase 2 – Data Collection & Preparation

- Collect raw data from chosen sources.
- Clean, filter, and format the data (handle missing values, duplicates, irrelevant features).
- Document challenges in data acquisition.
- Deliverable: Dataset (raw + cleaned) with description of preprocessing steps.

Phase 3 – Data Exploration & Feature Engineering

- Perform exploratory data analysis (EDA): descriptive stats, visualizations.
- Detect trends, patterns, and correlations.
- Engineer relevant features (transformations, new variables, encodings).
- Deliverable: EDA report with charts, key findings, and feature set description.

Phase 4 – Modeling & Analysis

- Choose suitable analytical or machine learning methods.
- Train and evaluate models (e.g., regression, classification, clustering, deep learning, etc.).
- Optimize parameters and compare results.
- Deliverable: Model results with performance metrics and justification of chosen approach.

Phase 5 – Insights, Visualization & Final Report

- Interpret results and link them to the original problem.
- Provide actionable insights or recommendations.
- Develop clear visualizations and dashboards if applicable.
- Deliverable: Final report (10–15 pages) + Presentation (10–12 minutes).

☞ By the end of the semester, each team will have walked through a **complete data analytics pipeline** from idea → data → model → insights.

Grading Breakdown Summary:

Phase 1: 10%

Phase 2: 15%

Phase 3: 15%

Phase 4: 20%

Phase 5: 30%

Overall Evaluation: 10%

= 100% (Project Weight = 20% of course)